

Introduction to Network Layer

Network Layer

Goals:

- ❑ understand principles behind network layer services:
 - routing (path selection)
 - dealing with scale
 - how a router works
 - advanced topics: IPv6, mobility, and queue management
- ❑ instantiation and implementation in the Internet

Chapter 4: Network Layer

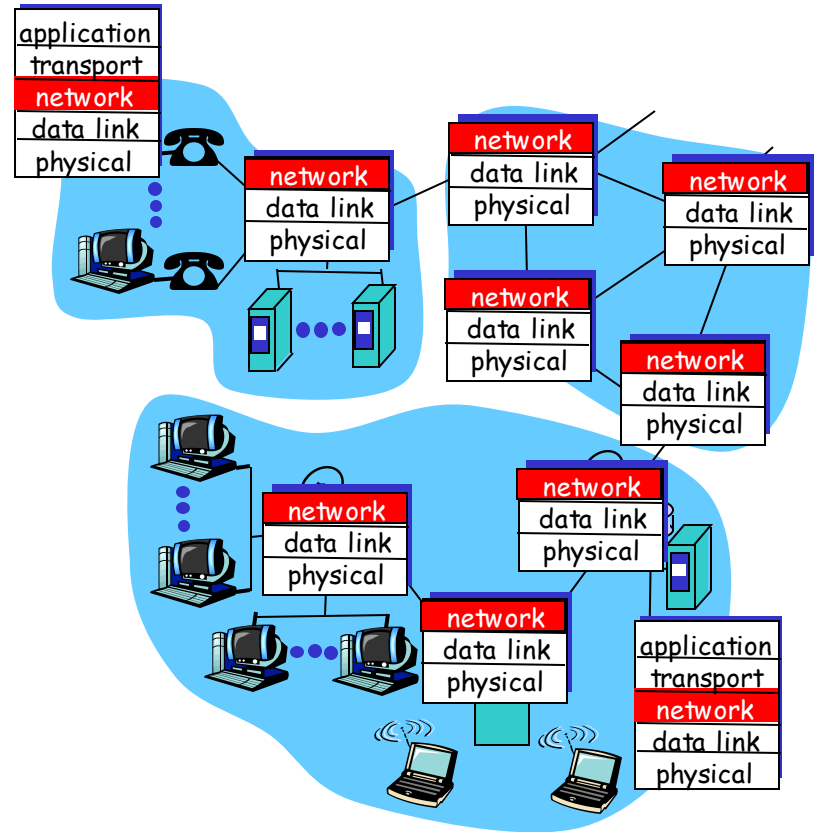
- ❑ 4.1 Introduction
- ❑ 4.2 Virtual circuit and datagram networks
- ❑ 4.3 What's inside a router
- ❑ 4.4 IP: Internet Protocol
 - Datagram format
 - IPv4 addressing
 - ICMP
 - IPv6
- ❑ 4.5 Routing algorithms
 - Link state
 - Distance Vector
 - Hierarchical routing
- ❑ 4.6 Routing in the Internet
 - RIP
 - OSPF
 - BGP
- ❑ 4.7 Broadcast and multicast routing

Network layer functions

- ❑ transport packet from sending to receiving hosts
- ❑ network layer protocols in every host, router

three important functions:

- ❑ *path determination*: route taken by packets from source to dest. *Routing algorithms*
- ❑ *forwarding*: move packets from router's input to appropriate router output
- ❑ *call setup*: some network architectures require router call setup along path before data flows



First Two Network-Layer Functions

□ *forwarding*: move packets from router's input to appropriate router output

□ *routing*: determine route taken by packets from source to dest.

- *Routing algorithms*

analogy:

□ *routing*: process of planning trip from source to dest

□ *forwarding*: process of getting through single interchange

The Third Function

- ❑ Connection setup in some network architectures:
 - ATM, frame relay, X.25
- ❑ Before datagrams flow, two hosts and intervening routers establish virtual connection
 - Routers get involved
- ❑ Network and transport layer cnctn service:
 - **Network:** between two hosts
 - **Transport:** between two processes

Network service model

Q: What *service model* for “channel” transporting datagrams from sender to rcvr?

Example services for individual datagrams:

- ❑ guaranteed (loss-less) delivery
- ❑ Guaranteed delivery with less than 40 msec delay (delay-conscious)

Example services for a flow of datagrams:

- ❑ In-order datagram delivery
- ❑ Guaranteed minimum bandwidth to flow
- ❑ Restrictions on changes in inter-packet spacing (no jitter)

The most important abstraction provided by network layer:
virtual circuit or datagram?

Network layer service models:

Network Architecture	Service Model	Guarantees ?				Congestion feedback
		Bandwidth	Loss	Order	Timing	
Internet	best effort	none	no	no	no	no (inferred via loss)
ATM	CBR	constant rate	yes	yes	yes	no congestion
ATM	VBR	guaranteed rate	yes	yes	yes	no congestion
ATM	ABR	guaranteed minimum	no	yes	no	yes
ATM	UBR	none	no	yes	no	no

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Network layer connection and connection-less service

- ❑ Datagram network provides network-layer connectionless service
- ❑ VC network provides network-layer connection service
- ❑ Analogous to the transport-layer services, but:
 - **Service:** host-to-host
 - **No choice:** network provides one or the other
 - **Implementation:** in the core

Virtual circuits

“source-to-dest path behaves much like telephone circuit”

- performance-wise
- network actions along source-to-dest path

- ❑ call setup, teardown for each call *before* data can flow
- ❑ each packet carries VC identifier (not destination host address)
- ❑ every router on source-dest path maintains “state” for each passing connection
- ❑ link, router resources (bandwidth, buffers) may be *allocated to VC*

VC implementation

A VC consists of:

1. Path from source to destination
2. VC numbers, one number for each link along path
3. Entries in forwarding tables in routers along path

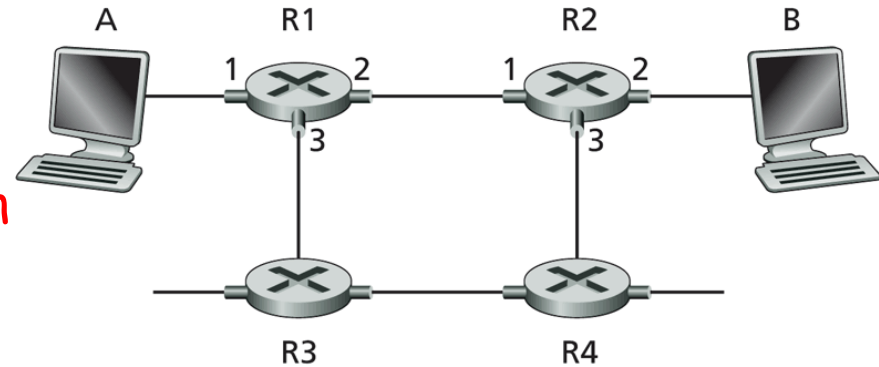
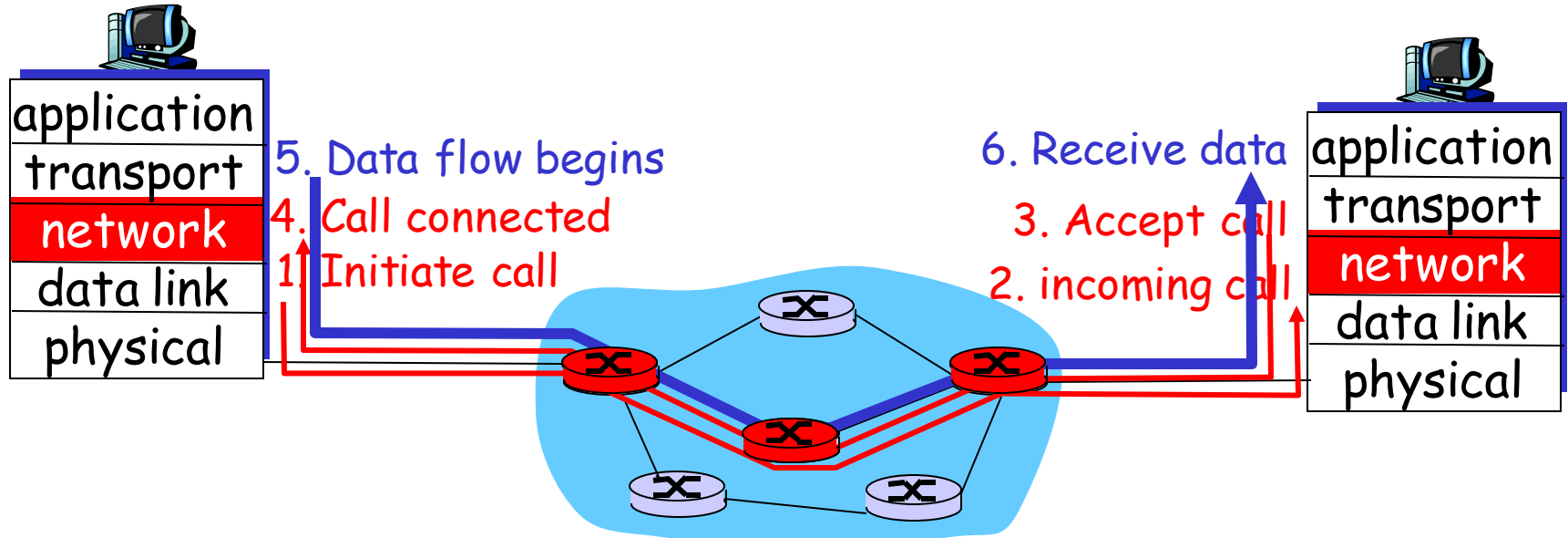


Figure 4.3 ♦ A simple virtual circuit network

- ❑ Packet belonging to VC carries a VC number.
- ❑ VC number must be changed on each link.
 - New VC number comes from forwarding table

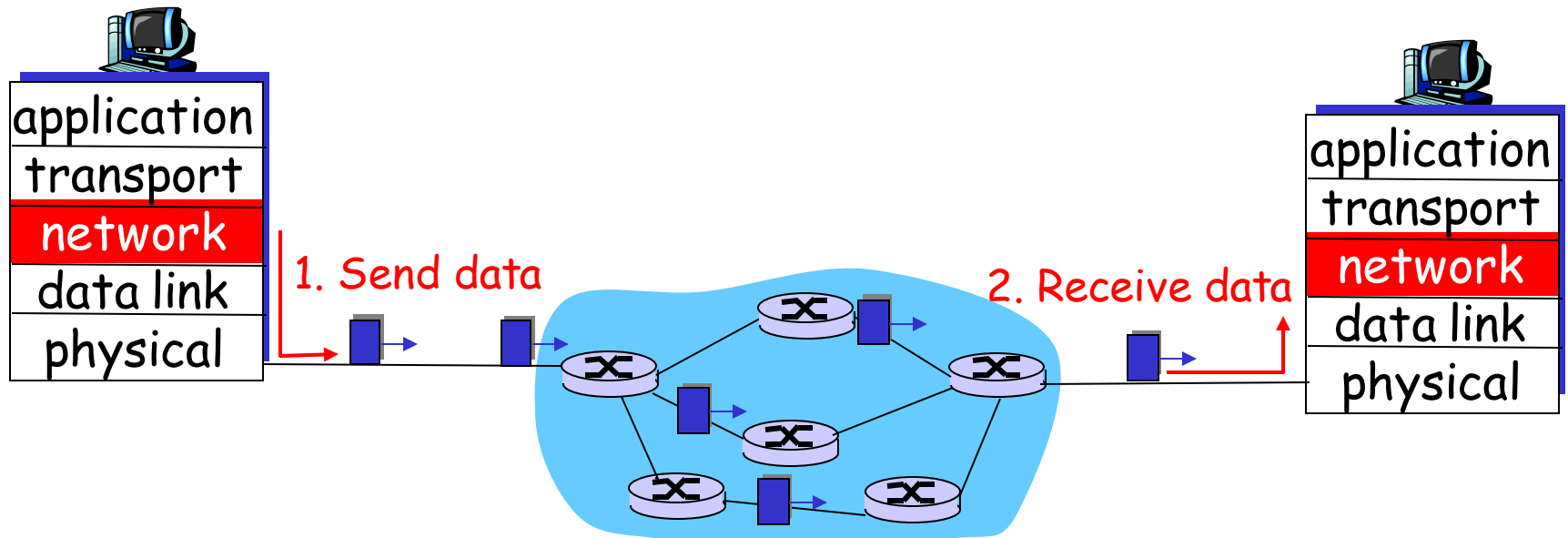
Virtual circuits: signaling protocols

- used to setup, maintain, and teardown VC
- used in ATM, frame-relay, X.25
- not used in today's Internet



Datagram networks: the Internet model

- ❑ no call setup at network layer
- ❑ routers: no state about end-to-end connections
 - no network-level concept of “connection”
- ❑ packets forwarded using destination host address
 - packets between same source-dest pair may take different paths



Forwarding table

4 billion
possible entries

<u>Destination Address Range</u>	<u>Link Interface</u>
11001000 00010111 00010000 00000000 through 11001000 00010111 00010111 11111111	0
11001000 00010111 00011000 00000000 through 11001000 00010111 00011000 11111111	1
11001000 00010111 00011001 00000000 through 11001000 00010111 00011111 11111111	2
otherwise	3

Longest prefix matching

<u>Prefix Match</u>	<u>Link Interface</u>
11001000 00010111 00010	0
11001000 00010111 00011000	1
11001000 00010111 00011	2
otherwise	3

Examples

DA: 11001000 00010111 00010110 10100001

Which interface?

DA: 11001000 00010111 00011000 10101010

Which interface?

Datagram or VC network: why?

Internet

- ❑ data exchange among computers
 - “elastic” service, no strict timing req.
- ❑ “smart” end systems (computers)
 - can adapt, perform control, error recovery
 - simple inside network, complexity at “edge”
- ❑ many link types
 - different characteristics
 - uniform service difficult

ATM

- ❑ evolved from telephony
- ❑ human conversation:
 - strict timing, reliability requirements
 - need for guaranteed service
- ❑ “dumb” end systems
 - telephones
 - complexity inside network